

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1712

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

*I **SADHANA S (192524298)** _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SADHANA S

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Smart Home Automation System Using Intelligent Agents

A smart home automation system controls **lighting, temperature, and security** using sensor data (motion, temperature, and time) while learning user preferences. Different intelligent agents can be used to achieve comfort, energy efficiency, and security.

(a) Simple Reflex Agent

Working

- Makes decisions based only on the current sensor input.
- Uses condition-action rules.

Example

- If motion is detected → Turn ON the lights.
- If no motion → Turn OFF the lights.

Advantages

- Fast response.
- Simple and easy to implement.

Disadvantages

- Cannot remember previous states.
- Cannot learn user preferences.

(b) Model-Based Agent

Working

- Maintains an internal model of the environment.
- Uses previous information along with current inputs.

Example

- Remembers that the living room light is already ON.
- Adjusts temperature based on previous room conditions.

Advantages

- Works in partially observable environments.
- Makes more accurate decisions.

Disadvantages

- Requires memory.
- More complex than reflex agents.

(c) Goal-Based Agent

Working

- Selects actions that achieve a predefined goal.

Example

Goals include:

- Maintain room temperature at **24°C**.
- Ensure all doors are locked at night.
- Turn OFF unused appliances.

Advantages

- Goal-oriented.
- Flexible decision-making.

Disadvantages

- May require additional computation.

(d) Utility-Based Agent

Working

- Chooses the action with the highest utility (overall benefit).
- Considers multiple factors simultaneously.

Example

The agent balances:

- User comfort
- Energy consumption
- Security
- Cost

If reducing the AC temperature increases electricity usage unnecessarily, the agent selects a more energy-efficient setting.

Advantages

- Produces the best overall decision.
- Handles trade-offs effectively.

Disadvantages

- Utility calculation is more complex.

Comparison

Agent Type	Memory	Goal-Oriented	Learning	Best Use
Simple Reflex	No	No	No	Basic lighting
Model-Based	Yes	No	Limited	Temperature control
Goal-Based	Yes	Yes	No	Security management
Utility-Based	Yes	Yes	Yes	Complete smart home automation

Best Choice

A Hybrid Agent (Model-Based + Utility-Based) is the most effective.
Justification

- Learns user preferences over time.
- Maintains an internal model of the home.
- Balances comfort, energy efficiency, and security.
- Makes intelligent decisions using real-time sensor data.
- Suitable for dynamic and partially observable environments.

Conclusion:

A Hybrid Agent provides the highest performance because it combines memory, reasoning, and utility optimization.

Question 2

Robot Navigation in an Unknown Maze

A robot must find the exit in an unknown maze without prior knowledge of the environment. Uninformed search strategies such as **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)** can be used.

(a) Uniform Cost Search (UCS)

Working

1. Start from the initial position.
2. Store nodes in a priority queue.
3. Expand the node with the lowest cumulative cost.
4. Continue until the goal is reached.
5. Returns the minimum-cost path.

Advantages

- Complete.
- Optimal.
- Suitable when movement costs vary.

Disadvantages

- Requires large memory.
- Slower for very large search spaces.

Complexity

- Time Complexity: $O(b^{(1+C/\epsilon)})^*$
- Space Complexity: $O(b^{(1+C/\epsilon)})^*$

Where:

- **b** = branching factor
- **C*** = optimal solution cost
- **ε** = minimum edge cost

(b) Iterative Deepening Search (IDS)

Working

1. Perform Depth-Limited Search with depth = 0.
2. If the goal is not found, increase the depth limit.
3. Repeat until the goal is reached.

Advantages

- Requires less memory.
- Complete.
- Suitable when solution depth is unknown.

Disadvantages

- Repeats node expansions.

- Not optimal when path costs differ.

Complexity

- Time Complexity: $O(b^d)$
- Space Complexity: $O(bd)$

Where:

- **b** = branching factor
- **d** = depth of the solution

Comparison of UCS and IDS

Feature	UCS	IDS
Search Type	Cost-based	Depth-based
Optimal	Yes	No (if costs vary)
Complete	Yes	Yes
Memory	High	Low
Best For	Weighted graphs	Unknown solution depth

Intelligent Agent for the Maze

The robot acts as a **Goal-Based Intelligent Agent**.

Characteristics

- Goal: Reach the exit safely.
- Perceives the environment using sensors.
- Chooses actions based on the current state.
- Updates decisions as new paths are discovered.

Agent Design and Decision-Making

The robot:

- Observes neighbouring cells.
- Avoids blocked paths.
- Selects the safest and least-cost route.
- Re-plans its path when obstacles are encountered.

A well-designed agent improves navigation accuracy, reduces unnecessary movements, and reaches the goal efficiently.

Best Combination

Goal-Based Agent + Uniform Cost Search (UCS)

Justification

- UCS guarantees the least-cost path.
- Goal-Based Agent focuses on reaching the exit.
- Suitable for unknown environments with varying movement costs.
- Supports intelligent decision-making based on current observations.
- Ensures efficient and reliable navigation.

Conclusion

Using a **Hybrid Goal-Based Intelligent Agent** with **Uniform Cost Search (UCS)** provides the best solution for robot navigation because it guarantees an optimal path, adapts to changing environments, and minimizes travel cost while successfully reaching the destination.